Experiment 6

Workflow Automation Layer for the Transport Management System

Aim

To design and build an automation layer in Zoho Creator for the Transport Management System, covering all eight Workflow options. Three workflows are implemented (Vehicle Status Update, Trip Completion, Insurance Expiry Alert); the others are planned.

Procedure

1. **Vehicle_Status_Update_Workflow (Form Workflow) [Implemented]:** when Trip Status becomes In Progress, the vehicle status changes to On Trip.
2. **Trip_Completion_Workflow (Form Workflow) [Implemented]:** when Trip Status becomes Completed, the vehicle status changes to Available.
3. **Vehicle_Insurance_Expiry_Schedule (Schedule) [Implemented]:** runs daily and emails the administrator when insurance is within 30 days of expiry.
4. **Trip_Request_Approval (Approval) [Planned]:** long-distance trips would need Transport Manager approval.
5. **Trip_Fare_Payment_Workflow (Payments) [Planned]:** would generate a payment link and update the payment status.
6. **Trip_Lifecycle_Blueprint (Blueprint) [Planned]:** restricts stage transitions to specific roles with an audit trail.
7. **Bulk_ExpiryAlert_Batch (Batch Workflow) [Planned]:** sends insurance and licence expiry alerts in batches.
8. **Generate_Code_Function (Function) [Planned]:** reusable Deluge function for VEH-001 style codes; a Report Workflow keeps grid edits in sync.

Workflow Summary

Workflow Type	Name	Trigger	Action
Form Workflow	Vehicle_Status_Update_Workflow; Trip_Completion_Workflow	On edit of Trip Status	Vehicle Status: On Trip / Available
Schedule	Vehicle_Insurance_Expiry_Schedule	Every day	Alert for insurance expiring in 30 days
Approval	Trip_Request_Approval (planned)	Distance above threshold	Manager approves or rejects
Payments	Trip_Fare_Payment_Workflow (planned)	On payment request	Payment link; update status
Blueprint	Trip_Lifecycle_Blueprint (planned)	Status transitions	Role-restricted stages, audit trail
Batch Workflow	Bulk_ExpiryAlert_Batch (planned)	Scheduled / on demand	Batch expiry alerts
Report Workflow	OnEdit_TripStatus_Report_Workflow (planned)	On edit in report	Keep vehicle status in sync
Function	Generate_Code_Function (planned)	Called by other workflows	Generate VEH-001 style codes

Sample Deluge: Form Workflows (On Success)

```
// Vehicle_Status_Update_Workflow
if(input.Trip_Status == "In Progress")
{
    vehicleRec = Vehicle_Details[ID == input.Vehicle];
    if(vehicleRec.count() > 0)
    {
        vehicleRec.Vehicle_Status = "On Trip";
    }
}
// Trip_Completion_Workflow
if(input.Trip_Status == "Completed")
{
    vehicleRec = Vehicle_Details[ID == input.Vehicle];
    if(vehicleRec.count() > 0)
    {
        vehicleRec.Vehicle_Status = "Available";
    }
}
```

Process Diagram

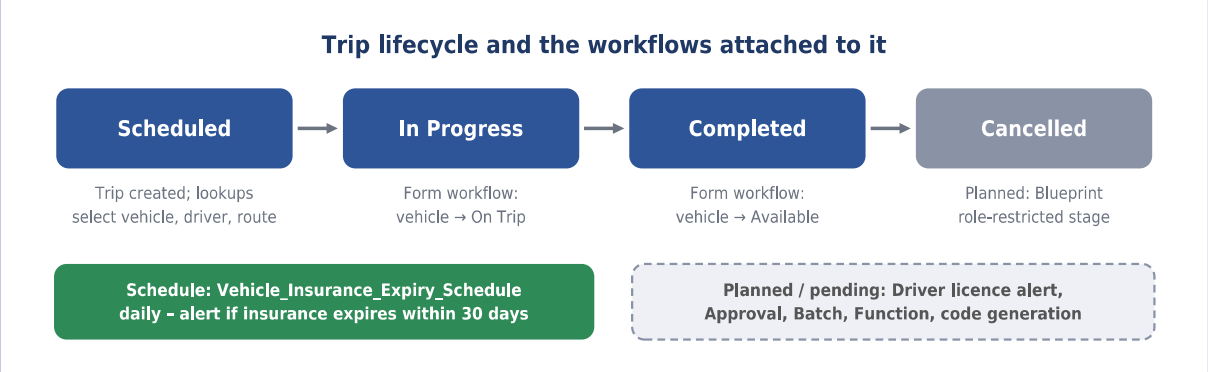


Figure 1: Diagram of the trip lifecycle and the workflows attached to it; blue/green = implemented, grey dashed = planned (illustrative diagram, not a screenshot).

Workflow Tests and Expected Outcome

All outcomes below are expected results; actual results are to be recorded after testing.

Workflow	Test Action	Expected Outcome
Vehicle_Status_Update_Workflow	Set Trip Status to In Progress	Selected vehicle's status changes to On Trip
Trip_Completion_Workflow	Set Trip Status to Completed	Selected vehicle's status changes to Available
Vehicle_Insurance_Expiry_Schedule	Vehicle with insurance expiring in 30 days exists	Alert email with vehicle number and expiry date is sent
Trip_Request_Approval (planned)	Trip above the distance threshold	Request goes to the Transport Manager for approval
Trip_Fare_Payment_Workflow (planned)	Payment requested	Payment link generated (needs a paid plan)
Trip_Lifecycle_Blueprint (planned)	Move a trip to the next stage	Only the permitted role can make the transition
Batch / Function (planned)	Run batch; add a record	Alerts sent in batches; next code generated

Sample Deluge: Insurance Expiry Schedule

```
expiring = Vehicle_Details[Insurance_Expiry_Date <= zoho.currentdate.addDay(30)];
for each v in expiring
{
    sendmail
    [
        from : zoho.adminuserid
        to : zoho.adminuserid
        subject : "Insurance Expiry Alert - " + v.Vehicle_Number
        message : "Vehicle " + v.Vehicle_Number + " insurance expires on "
                  + v.Insurance_Expiry_Date + ". Please renew it."
    ]
}
```

Blueprint Transitions and Roles (Planned)

From Stage	To Stage	Allowed Role
Scheduled	In Progress	Transport Coordinator
In Progress	Completed	Driver / Transport Coordinator
Scheduled	Cancelled	Transport Manager
In Progress	Cancelled	Transport Manager

Insurance Alert Email (Expected)

```
Subject: Insurance Expiry Alert - <Vehicle_Number>

Dear Administrator,
The insurance of vehicle <Vehicle_Number> expires on <Insurance_Expiry_Date>.
Please renew it before the expiry date.

Regards,
Transport Management System
```

Implementation Status

Workflow	Type	Status
Vehicle Status Update; Trip Completion	Form Workflow	Implemented
Vehicle Insurance Expiry Alert	Schedule	Implemented
Driver Licence Expiry Alert	Schedule	Planned / pending
Auto codes: VEH-001, DRV-001, PAS-001, RTE-001, TRP-001	Form Workflow / Function	Planned / pending
Approval, Payments, Blueprint, Batch, Report Workflow	Design only	Planned / pending

Result

The automation layer for the Transport Management System was designed covering all eight Workflow options. The Vehicle Status Update, Trip Completion and Vehicle Insurance Expiry Alert workflows are implemented; the remaining workflows are planned.